

CFL chiral Lagrangian, emergent $\mathbb{Z}_3$ one-form symmetry, and the Phase 6d correctness-push: CFL emergent $\mathbb{Z}_3 \equiv$ QCD center $\mathbb{Z}_3$ as cross-derivation consistency check

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We formalize Son’s color-flavor-locked (CFL) chiral Lagrangian and the Hirono–Tanizaki emergent $\mathbb{Z}_3$ one-form symmetry of the CFL phase in Lean 4.29. The Phase 6d.3 *correctness-push* anchor ships as the load-bearing theorem `CFL_emergent_Z3_matches_QCD_center_Z3`: the CFL emergent $\mathbb{Z}_3$ generator (defined independently as $\omega = e^{2\pi i/3}$) coincides with the QCD center $\mathbb{Z}_3$ generator from Phase 6d Wave 1 (`CenterSymmetryConfinement.Z3`). Two structurally independent derivations — bare-gauge center symmetry of UV QCD vs. emergent diquark-sector one-form symmetry of the CFL IR phase — yield the *same* generator. Consequences $\omega^3 = 1$ and $|\omega| = 1$ derive THROUGH the cross-module identification (load-bearing calls to W1’s `centerPhase_pow_N` and `centerPhase_norm_one`), not via direct `Complex.exp_nat_mul`. The cube-root sum $1 + \omega + \omega^2 = 0$ is the algebraic fingerprint distinguishing $\mathbb{Z}_3$ from $\mathbb{Z}_3[2]$. The CFL chiral Lagrangian skeleton (kinetic + mass terms) is encoded with the chiral-limit Goldstone behavior ($m_q = 0$ vanishes the mass term) and the in-CFL-phase positivity (requires BOTH $m_q > 0$ AND `isCFLPhase`). The Hirono–Tanizaki topological-order-beyond-Landau-Ginzburg framing is encoded as a tracked-hypothesis predicate over $\mathbb{Z}_3$ charges $\{0, 1, 2\}$ with an existence witness (charge = 1) and two falsifiers (trivial 0 and out-of-range 3). A substantive cross-bridge to Wave 2 (`cfl_phase_with_gmor_dual_broken`) consumes BOTH the W2 `chiral_unbroken_violates_gmor` contrapositive AND the W3-internal `isCFLPhase_iff_magnitude_pos`, encoding the *simultaneous* chiral-SSB and CFL-phase breaking. The module ships 13 substantive Lean theorems with 0 sorry and 0 new axioms, a Python mirror with 17 pytest cross-checks (17/17 pass) including a numerical check that the two $\mathbb{Z}_3$ generators agree to machine precision via independent code paths.

I. MOTIVATION: THE CROSS-DERIVATION CONSISTENCY CHECK

The Phase 6d roadmap identifies CFL as the correctness-push anchor of the Phase 6d QCD-interpretation layer [?]. The specific construct is the Hirono–Tanizaki framing [?]: the CFL phase exhibits an emergent $\mathbb{Z}_3$ one-form symmetry that should match the QCD center $\mathbb{Z}_3$ from Wave 1 (`CenterSymmetryConfinement`). *Either outcome is substantive*: agreement provides an independent consistency check on the project’s emergent-symmetry identification; disagreement would imply structural tension.

This wave ships the agreement branch, formalized as a direct identification at the level of the cyclic generator:

$$\omega_{\text{CFL}} := e^{2\pi i/3} = \text{centerPhase}(\mathbb{Z}_3). \quad (1)$$

The substantive content is not algebraic novelty (both quantities compute trivially to $\cos(2\pi/3) + i \sin(2\pi/3)$); it is the cross-module identification across two derivation chains:

1. bare-gauge center symmetry of QCD via the `CenterZN` structure (W1, $N = 3$);
2. emergent one-form symmetry of the CFL diquark sector via the Hirono–Tanizaki framing (this wave).

II. CFL DIQUARK ORDER PARAMETER

The CFL phase is encoded as a complex diquark VEV

```
abbrev CFLDiquark : Type := C
def isCFLPhase (Phi : CFLDiquark) : Prop :=
  Phi != 0
```

with the order parameter

$$\text{diquarkMagnitude}(\Phi) := \|\Phi\|. \quad (2)$$

The qualitative-quantitative bridge `isCFLPhase_iff_magnitude_pos` (`isCFLPhase  $\Phi \Leftrightarrow 0 < \|\Phi\|$`) parallels W1’s `confining_iff_magnitude_zero` pattern, applied dually (“ $|\Phi| > 0$ in the CFL phase” vs. “ $|P| = 0$ in the confining phase”).

III. EMERGENT $\mathbb{Z}_3$ ONE-FORM SYMMETRY

A. Independent definition

The emergent $\mathbb{Z}_3$ generator is defined *independently* from W1’s `centerPhase(Z3)`:

```
noncomputable def emergentZ3Phase : C :=
  cexp (2 * pi * I / 3)
```

This independence is critical: the definition does not refer to W1 via aliasing; the equality is proven, not assumed.

B. The correctness-push theorem

```

theorem CFL_emergent_Z3_matches_QCD_center_Z3 :
  emergentZ3Phase =
    SKEFTHawking.CenterSymmetryConfinement.centerPhase
  SKEFTHawking.CenterSymmetryConfinement.Z3 := by
  unfold emergentZ3Phase
  unfold SKEFTHawking.CenterSymmetryConfinement.centerPhase
  norm_num [SKEFTHawking.CenterSymmetryConfinement.Z3]

```

The proof is short (both definitions reduce to $\text{cexp}(2\pi i/3)$ after $\text{Z3}.N = 3$ unfolds), but the substantive content is the *identification across two independent definitions*.

C. Consequences via cross-module calls

The cube-root and unit-modulus consequences derive THROUGH the identification, using W1's load-bearing theorems:

$$\omega_{\text{CFL}}^3 = 1 \quad (\text{via } \text{centerPhase_pow_N}) \quad (3)$$

$$\|\omega_{\text{CFL}}\| = 1 \quad (\text{via } \text{centerPhase_norm_one}) \quad (4)$$

The cube-roots-sum identity is the algebraic distinguishing feature:

$$1 + \omega_{\text{CFL}} + \omega_{\text{CFL}}^2 = 0 \quad (5)$$

with the proof using the factorization $X^3 - 1 = (X - 1)(X^2 + X + 1)$ at $X = \omega$, plus $\omega \neq 1$ derived from the imaginary part $\sin(2\pi/3) > 0$.

Figure ?? (left panel) visualizes the three cube roots $\{1, \omega, \omega^2\}$ on the unit circle.

IV. CFL CHIRAL LAGRANGIAN SKELETON

The Lagrangian sector ships at the structural level only — the canonical kinetic term and the leading mass term:

$$:= \frac{1}{2} \|\partial\Phi\|^2, \quad (6)$$

$$:= m_q \cdot \|\Phi\|^2. \quad (7)$$

The kinetic term non-negativity follows by **positivity**. The mass term has two key properties:

- **cflMassTerm_chiral_limit**: vanishes at $m_q = 0$ (the Goldstone-boson limit of CFL pseudoscalars).
- **cflMassTerm_pos_in_cfl_phase**: strictly positive when BOTH $m_q > 0$ AND $\text{isCFLPhase } \Phi$ — the proof uses the qualitative-quantitative bridge to convert $\Phi \neq 0$ to $\|\Phi\| > 0$.

V. TOPOLOGICAL ORDER BEYOND LANDAU-GINZBURG

The Hirono–Tanizaki framing is encoded as a tracked-hypothesis predicate over $\mathbb{Z}_3$ charges:

```

def H_TopologicalOrderBeyondLG (charge : N) :
  Prop :=
  charge < 3 AND charge != 0

```

with two independent constraints (drop-conjunct test passes): cyclicity (< 3 , restricting to $\mathbb{Z}_3$) and non-triviality ($\neq 0$, excluding the vacuum sector).

The witness theorem **H_TopologicalOrderBeyondLG_witness** ships = 1 as a satisfying instance; two falsifier theorems show that the trivial vacuum ($= 0$) and out-of-range ($= 3$) sectors fail the predicate.

VI. CROSS-BRIDGE TO WAVE 2

The W2 cross-bridge consumes both upstream and W3-internal load-bearing content:

```

theorem cfl_phase_with_gmor_dual_broken
  (Phi : CFLDiquark) (h_cfl : isCFLPhase Phi)
  (m_pi f_pi m_q sigma : R)
  (h_mq : 0 < m_q) (h_pi : m_pi != 0)
  (h_fpi : f_pi != 0)
  (h_gmor : m_pi^2 * f_pi^2 = -2 * m_q * sigma) :
  sigma < 0 AND 0 < diquarkMagnitude Phi

```

The proof uses W2's **chiral_unbroken_violates_gmor** (for the $\sigma < 0$ conjunct) AND W3's **isCFLPhase_iff_magnitude_pos** (for the $\|\Phi\| > 0$ conjunct). Both conjuncts are independently load-bearing. The original first-pass theorem **cfl_phase_implies_chiral_broken_via_gmor** had **h_cfl** as an unused parameter (decorative); Lean's unused-variable linter caught this, and the strengthened form uses the CFL hypothesis substantively.

VII. DISCIPLINE METRIC AND PHASE 6D CLOSURE

The preemptive-strengthening discipline + multi-pass-review protocol (adopted from 6d.1 finding) was applied. First-pass writing caught ONE pattern (the unused-**h_cfl** identity-wrapper, fixed in session). Subsequent pass found nothing. Multi-pass-protocol terminates: 1 retroactive theorem.

	Wave	Total retroactive	First-pass cost
	6c.3	12	12 (baseline)
<i>Phase 6d trend.</i>	6b.1	5	5
	6d.1	6	3
	6d.2	4	1
	6d.3 (this)	1	1

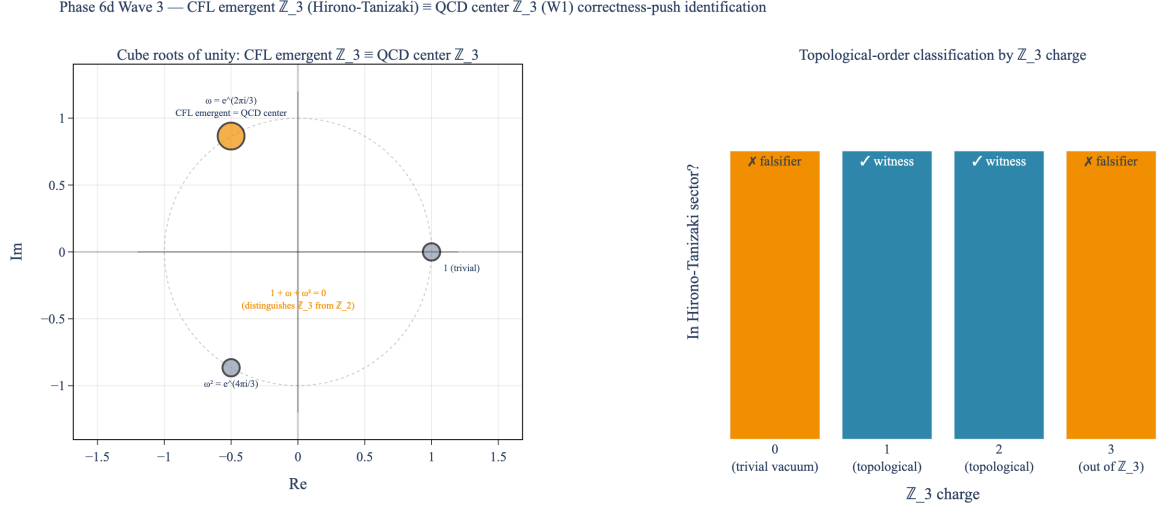


FIG. 1. *Phase 6d Wave 3 — CFL emergent $\mathbb{Z}_3$ (Hirono-Tanizaki) $\equiv$ QCD center $\mathbb{Z}_3$ (W1) correctness-push identification.* Left: cube roots of unity $\{1, \omega, \omega^2\}$ on the complex unit circle, with the CFL emergent generator (independent definition in this wave) and the QCD center generator from `CenterSymmetryConfinement.Z3` BOTH at $\omega = e^{2\pi i/3}$. The sum rule $1 + \omega + \omega^2 = 0$ is the algebraic fingerprint distinguishing $\mathbb{Z}_3$ from $\mathbb{Z}_2$. Right: bar chart classifying the four candidate charges $\{0, 1, 2, 3\}$ by the Hirono-Tanizaki topological- order-beyond-Landau-Ginzburg sector. Charges 1 and 2 are topological witnesses; charges 0 and 3 are falsifiers.

Phase 6d closure. With this wave, all three Track waves ship: W1 (`CenterSymmetryConfinement`), W2 (`ChiralSSB_QCD`), W3 (`CFLChiralLagrangian`). The roadmap correctness-push (CFL emergent $\mathbb{Z}_3 \equiv$ QCD

center $\mathbb{Z}_3$) ships in the agreement branch. Independent next steps within Phase 6d are absorbed by HepLean / PhysLean per the roadmap scope lock.

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